

Nova AI

Reusable AI Starter Template

A beginner-friendly, production-quality web application that uses Google Gemini to power chat and many AI product ideas from one reusable codebase.

Project Name	Nova AI — AI Starter Template
Type	Full-Stack AI Web Application (Frontend + Gemini API)
Stack	React, Vite, JavaScript, Tailwind CSS, Google Gemini
Purpose	Hackathon / Portfolio / Reusable AI Product Foundation
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2. Introduction

Nova AI is a modern AI starter template. It looks and feels like a premium SaaS product (similar to ChatGPT-style apps), but the code is designed so you can reuse it for many different AI tools.

Instead of building a new project every time, you can keep the same architecture and only change prompts and UI text to create products such as:

- AI Chatbot
- AI Resume Analyzer
- AI Study Assistant
- AI Interview Coach
- AI Career Advisor
- AI Code Reviewer
- AI Email Assistant
- AI Project Generator
- AI Research Assistant

3. Project Objectives

Beginner Friendly
Clear folders, simple JavaScript, easy setup with npm.

Production Quality
Clean UI, reusable components, and proper error handling.

Highly Reusable
One Gemini service + prompt system for many AI apps.

Portfolio Worthy
Premium glassmorphism UI suitable for demos and recruiters.

4. Technology Stack

Layer	Technology	Why it is used
UI Framework	React (Vite)	Fast development and modern component-based UI
Language	JavaScript	Easy for beginners; no TypeScript complexity
Styling	Tailwind CSS	Quick premium styling with consistent design tokens
Icons	React Icons	Clean icons for navigation and actions
Markdown	React Markdown + remark-gfm	Beautiful AI answers with lists, tables, and formatting
Code Highlight	Highlight.js	Readable code blocks in AI responses
AI Engine	Google Gemini (`@google/genai`)	Latest official SDK for generating AI replies
Routing	React Router	Landing page and Chat page navigation

5. System Architecture

The project follows a clean layered structure:

Folder	Responsibility
src/pages	Full screens: Landing Page and Chat Page
src/layouts	Page shells (landing shell and app shell with sidebars)
src/components	Reusable UI pieces (chat bubbles, buttons, cards)
src/context	Shared app state: theme, chat sessions, toasts
src/hooks	Logic helpers like send message and auto-scroll
src/services	Only place that talks to Gemini API
src/constants	App name, prompts, templates
src/utils	Error mapping and small helper functions

This separation makes the project easy to understand, maintain, and extend.

6. Pages and Layout

6.1 Landing Page

The first page users see. It is designed like a premium AI SaaS dashboard and includes:

- Brand/hero panel with project name and key benefits
- Studio preview showing how the chat product looks
- Quick template shortcuts
- Usage overview (demo stats)
- Product cards for many AI use cases
- Footer strip with value points

6.2 Chat Studio Page (/chat)

The main working screen of the system. It includes:

- **Left Sidebar:** New Chat button and conversation history
- **Center Chat Area:** messages, typing indicator, and input box
- **Right Rail:** Quick Templates and usage overview
- **Header:** theme toggle, clear chat, new chat

7. Feature Guide (All Features Explained)

This section explains every major feature in simple language.

7.1 Premium SaaS User Interface

The interface uses glass panels, soft shadows, rounded cards, gradient buttons, and smooth animations. The goal is to look like modern products such as ChatGPT, Linear, Notion AI, and Vercel dashboards.

7.2 Dark Mode and Light Mode

Users can switch between dark and light themes with one click. The selected theme is saved in browser storage, so it stays the same after refresh.

7.3 Landing Page Experience

The landing page introduces the product and shows what users can build. Buttons like **New Chat**, **Send**, and **Try Template** take the user into the real chat studio.

7.4 Sidebar Navigation

The left sidebar helps users manage chats. They can start a new chat, open an old chat, or delete a chat. On mobile, the sidebar opens as a drawer.

7.5 Session Conversation History

Every conversation is saved as a session. Sessions are stored in `localStorage`, so history remains after closing and reopening the browser (on the same device).

7.6 New Chat

Creates a fresh empty conversation. Useful when switching topics so the AI does not mix old context with a new task.

7.7 Clear Chat

Removes messages from the current conversation while keeping the session entry. Helpful for resetting a chat quickly.

7.8 Chat Conversation Area

Shows the full message thread. User messages appear on the right. AI messages appear on the left with a sparkle avatar.

7.9 User Message Bubble

Displays what the user typed in a clean right-aligned bubble.

7.10 AI Message Bubble

Displays Gemini replies in a glass-style bubble. Supports rich formatting for easier reading.

7.11 Markdown Support

AI answers can include headings, bold text, lists, links, tables, and quotes. This makes long answers much easier to scan.

7.12 Code Syntax Highlighting

When the AI returns code, it is shown in a highlighted code block with language label and a copy button. Languages like JavaScript, Python, HTML, CSS, JSON, and more are supported.

7.13 Typing Indicator

While waiting for Gemini, animated dots appear so the user knows the system is working.

7.14 Loading Spinner

A spinner appears on the Send button during request processing and during route loading.

7.15 Auto Scroll

The chat automatically scrolls to the newest message so users always see the latest reply.

7.16 Enter to Send / Shift+Enter for New Line

Enter sends the message. **Shift + Enter** adds a new line inside the input box. This matches modern chat product behavior.

7.17 Copy AI Response

Users can copy a full AI reply (or a code block) to the clipboard with one click. A toast confirms the copy action.

7.18 Empty Chat Starter Prompts

When a chat is empty, starter cards appear (for example: explain a concept, improve a resume bullet, mock interview, code review). Clicking a card sends that prompt immediately.

7.19 Quick Templates (Right Panel)

The right rail provides ready templates such as Resume Analyzer, Study Assistant, Interview Coach, Code Reviewer, Email Assistant, and Research Assistant. Selecting one starts a guided prompt in chat.

7.20 Product Template Cards (Landing)

The landing page shows multiple AI product cards. Each card has an icon, short description, mini preview, and a "Try Template" button that opens the studio.

7.21 Toast Notifications

Small popup messages appear for success, info, warning, and error events (for example: "Chat cleared", "Response copied", API errors).

7.22 Error UI

Important problems are shown in a red error banner inside the chat area. Users can dismiss the banner after reading it.

7.23 Responsive Design

The layout works on desktop, tablet, and mobile. Side panels become drawers on smaller screens so the chat remains usable.

7.24 Environment-Based API Key

The Gemini API key is stored in a `.env` file as `VITE_GEMINI_API_KEY`. Secrets are not hardcoded in source files. `.env` is gitignored for safety.

7.25 Reusable Gemini Service

All AI calls go through `src/services/geminiService.js`. Pages and components do not talk to Gemini directly. This keeps the code clean and reusable.

7.26 Reusable Prompt System

System instructions and product presets live in `src/constants/prompts.js`. Changing prompts can turn the same app into a different AI product without rewriting the whole project.

8. How the AI Works (Simple Flow)

- 1 User types a message and presses Enter (or Send).
- 2 App checks that the message is not empty and that an API key exists.
- 3 User message is added to the active chat session.
- 4 `geminiService` sends chat history + new message + system prompt to Gemini.
- 5 Gemini returns a text reply.
- 6 App shows the reply with markdown/code formatting and saves it in session history.

9. Reusable Prompt System

The project includes product presets such as:

Preset	What it becomes
<code>chatbot</code>	General AI assistant

Preset	What it becomes
<code>resumeAnalyzer</code>	Resume feedback and rewrite helper
<code>studyAssistant</code>	Learning coach with simple explanations
<code>interviewCoach</code>	Mock interview practice
<code>careerAdvisor</code>	Career planning guidance
<code>codeReviewer</code>	Code quality and bug review
<code>emailAssistant</code>	Professional email drafting
<code>projectGenerator</code>	MVP and project idea planning
<code>researchAssistant</code>	Topic summaries and research help

To specialize the app, update branding in `appConfig.js` / `.env` and switch the active system prompt.

10. Error Handling and Safety

Problem	What the user sees
Empty message	Warning toast; no API call is made
Missing API key	Error banner with setup instructions
Network failure	Clear network error message
Invalid/unauthorized key	Actionable API key error message
Rate limit / quota	Retry later / change model guidance
Unavailable model	Message to update <code>VITE_GEMINI_MODEL</code>
Unexpected error	Safe fallback error message

Note: Because this is a frontend Vite app, `VITE_*` keys are visible in the browser. This is fine for demos and hackathons. For real production apps, API calls should go through a backend proxy.

11. How to Run the Project

- 1 Install packages: `npm install`
- 2 Create `.env` from `.env.example` and add your Gemini API key

- 3 Start development server: `npm run dev`
- 4 Open the local URL (usually `http://localhost:5173`)
- 5 Build for production: `npm run build`

12. Deployment

The project is ready for:

- **GitHub:** push source code (never commit `.env`)
- **Vercel:** import repository, set env vars, deploy as Vite app

`vercel.json` includes SPA rewrite rules so routes like `/chat` work correctly after deploy.

13. Future Improvements

- Backend proxy so API keys stay private
- Streaming responses (token-by-token output)
- File upload for resume/PDF analysis
- User authentication and cloud-synced history
- Voice input and richer multimodal features
- Prompt evaluation tools for quality testing

14. Conclusion

Nova AI is more than a chatbot demo. It is a reusable foundation for building many AI products quickly. With a clean architecture, a polished SaaS interface, Gemini integration, session history, markdown/code support, and a flexible prompt system, it is suitable for hackathons, learning, and portfolio presentation.

By changing prompts and UI branding, the same project can become a resume analyzer, interview coach, study assistant, or another specialized AI tool — without rebuilding everything from scratch.